# The Palaeomagnetism of the Bloomsburg Red Beds and its Possible Application to the Tectonic History of the Appalachians

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*Summary*

Thermal analysis of the fossil magnetism of the Silurian Bloomsburg red beds of Pennsylvania indicates that there are two magnetic components; a *thermally-discrete component* of very great stability which is unchanged in temperatures up to 650 °C and which disappears between 650 and 700 °C, and a *thermally-distributed component* whose properties are defined by a series of blocking temperatures mostly in the range 300 to 550 °C. The latter is roughly parallel to the supposed direction of the Permian geomagnetic field, and is considered to have been induced by moderate heating during burial at about the time of the Appalachian orogeny. The former has apparently been unaffected by these events and is considered to be of Silurian age. The discrete component has a mean declination 6° W of N diverging by $29\frac{1}{2}°$ from that observed by Graham (1949) in the Silurian Rosehill Formation which is situated some 300 km to the SW. The divergence is equal to the angular change in tectonic trend between the two areas. This result may be explained by supposing that this part of the Appalachian geosyncline was straight in Silurian time and that the change in trend is a subsequent strain; that is, the Appalachians at this point are an orocline (Carey 1958). This is not to say that alternative hypotheses will not explain the observations, but it is clear that the question can be decided by further palaeomagnetic studies along this mountain chain.

## 1. The tectonic problem

Orogenic belts are often curved in plan. Without attempting detailed review, explanations that have been put forward to explain this curvature may be conveniently considered under two heads.

### 1.1. *Primary origin*

These ideas, which have been developed for instance by Wilson (1954), are an extension of the contraction theory of orogenesis (Jeffreys 1959) which supposes that in a cooling Earth the outer 70 km or so is in compression due to shrinkage

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of the deeper parts. Failure in the outer shell may occur along conical zones (Scheiddeger 1958) which have arcuate surface traces marking the sites of geosynclines and subsequent mountain chains.

1.2. *Secondary origin*

This is the hypothesis of Carey (1958) who supposes that the curvature arises from bending in plan at a late stage in the history of the geosyncline; these bends he calls *oroclines* and the larger arcs, such as the Alaskan oroclines, are considered to have arisen as a result of relative movements between continental blocks.

The ideas can be tested by palaeomagnetism. The palaeomagnetic criterion for an orocline is that in beds of comparable age the horizontal component of magnetization makes the same angle with the trend in all parts of the arc, whereas the inclination is constant throughout. In an arc whose curvature is primary, both declination and inclination will remain constant irrespective of the changes in trend. This test does not depend on assumptions about the configuration of the palaeogeomagnetic field since, in a particular instance, the region concerned only forms a small portion of the Earth's surface. Kawai, Ito & Kume (1961) have presented data relevant to the bending of the Japanese arc, which is currently active. In this paper observations are given, which, in conjunction with previous data of Graham (1949), suggest a similar effect for part of the Appalachian mountain chain. This mountain chain, which is the site of the type geosyncline of Hall and Dana, received sediment during much of the Palaeozoic. Several orogenic phases are present, the final one being the Appalachian orogeny at the close of the Palaeozoic.

## 2. Geology and sampling

The Bloomsburg red beds are of Upper Silurian age and are situated in the Valley and Ridge Province of the Appalachians (Figure 11). Samples were taken from 18 levels spaced at irregular intervals through a section, about 85 m thick, exposed along a disused railway cutting at Fishing Creek near Bloomsburg, Pa. (41° 01′ N, 74° 27′ W). Results from the various levels are labelled 1 to 18. In outcrop the formation is well stratified with little variation in dip, the mean being 25° at 158° true bearing. On closer study in hand-sample and thin section, the bedding is seen to be highly irregular. Layers, a few millimetres thick and coloured a dark red, are contorted into folds and random whorls with dimensions ranging down from a centimetre. This deformation, which does not affect the main lie of the beds arises from slumping or from reworking by animals prior to lithification. Twenty-eight oriented hand-samples were obtained, either 1 or 2 at each level. The thickness spanned at each level varies from 5 to 30 cm. Disk specimens (74 in all), 3·5 cm in diameter and 0·7 cm thick, were cut from the samples.

Examination of thin sections shows that the rock is made up of clastic particles (predominantly quartz) in the grain-size range 10 to 100 $\mu$ (coarse silt and very fine sand) together with much fine grained interstitial material permeated by a red stain material of extremely fine grain-size. There are also occasional black crystalline particles with diameters mostly about 20 $\mu$. The red staining may be hematite or an oxyhydroxide of hematite. The black crystalline material is birefflective and is considered to be specular hematite. Van Houten (1961) has determined the black iron oxide grains in other Silurian red bed formations in this region as specular hematite.

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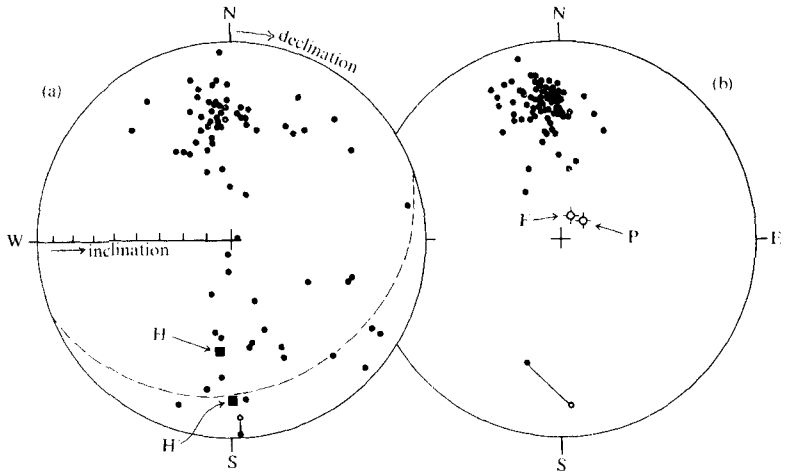



FIG. 1.—Directions of magnetization of rocks disks; (a) NRM, (b) after thermal cleaning at 550 °C. Equal-area polar projections are used the primitive being the bedding plane. Dots (circles) indicate negative (positive) inclinations. The dotted line in (a) is the trace of the horizontal plane on the upper hemisphere. The stars indicate the directions (positive inclinations) of the present geomagnetic field (*F*) and the geocentric axial dipole field (*P*). Disks from level 9 (*Type C*) are linked. The squares give the direction (negative inclination) of the supposed Permian field assuming beds folded prior to (*H*) and after ($H^1$) the Permian.

## 3. Natural remanent magnetization (NRM)

The directions and intensity of NRM are given in Figures 1(a) and 6(a) and (b). The uncertainty in the direction determinations (combined orientation and measurement error) is 3°, and the intensities are correct within 5 per cent. The directions fall into a northerly group with inclinations averaging about −30°, and a widely scattered group with NE, SE, and S declinations. The northerly group is characterized by good agreement between disks from the same sample (Figure 4), and between different samples. The scatter in the second group is due to divergences between samples and between disks from the same sample; an example of the latter effect is given in Figure 5 in which the differences range up to 130°. In all cases there is strong divergence (>80°) from the present geomagnetic field, and repeat measurements on disks which have been stored at random in the Earth's field for a year agree with the initial determinations within the accuracy of the measurements. There is no smearing towards the present field direction. These facts show that the rocks possess an appreciable magnetic stability, and have been little affected by the Earth's field over the past few thousand years.

## 4. Thermal demagnetization methods

Demagnetization techniques have been used to investigate the magnetic components present, and, in particular, to study the cause of the wide scatter. Demagnetization by alternating magnetic fields has not yielded useful results; the intensity and directions of remanence are so stable that they are unaffected in fields of peak amplitude 1000 Oe. A characteristic demagnetization curve exhibiting this behaviour has been given by Opdyke (1961). However, thermal demagnetization, of the type introduced by Thellier (1938), and subsequently applied to the study

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of the remanent magnetization of sediments by several workers (Leng 1955, Doell 1956, Irving & others 1961a, Chamalaun & Creer 1963) has proved very useful. The rock disks are heated to a given temperature, held there for 5 to 10 min., and then cooled in a space in which the geomagnetic field is compensated to about 0·2 per cent. The disk is measured immediately and then reheated to a higher temperature, cooled, and measured again. Details of the apparatus and procedures have been given previously (Irving & others 1961a). The effect of residual fields in the apparatus was minimized by taking several disks from each sample, and placing these in random positions in the furnace. The agreement between disks is good (Figures 1(b) and 4) suggesting that the effect of errors in field cancellation are negligible in the temperature ranges up to the Curie temperature, $T_c$. After cooling from above $T_c$ however, residual fields do become important, the observed remanence never in fact being zero; statistical fluctuations and chemical changes may also contribute to this "minimum intensity" (Irving & others 1961b).

Heating and cooling were carried out *in air*, which gave values agreeing well upon repeat heatings to the same temperature in the range up to 650 °C, showing that no magnetically appreciable chemical changes occur. Trial runs using a nitrogen atmosphere induced chemical changes (presumably reduction of hematite to magnetite) at about 500 °C as has previously been found in red sediments by Leng (1955) so that this is an undesirable procedure.

Thermal demagnetization may be thought of as the converse of the acquisition of thermoremanent magnetization (TRM). Above $T_c$ thermal agitations exceed the magnetic energy and the material is paramagnetic. As the temperature falls through $T_c$ the material acquires an equilibrium magnetization which moves freely into the direction of an applied field. At a certain temperature, called the blocking temperature $T_b$, this magnetization is frozen and can resist changes in the field applied on a short time scale (Neel 1955). In thermal demagnetization, as the temperature rises above $T_b$ for a given component, the magnetization of that component is randomized, and, in the absence of a magnetic field is not reacquired upon cooling, so that components with successively higher blocking temperatures are progressively eliminated.

A useful concept, derived from this theory, is that of thermally-discrete and thermally-distributed components. A *thermally-discrete component* is defined as a component in which the spectrum of blocking temperatures is narrow, being concentrated in a small range below $T_c$, which, for the purposes of these experiments, is taken as $<50$ °C. The demagnetization curve is of rectangular form. The example given here (Figure 2) is from Carboniferous red beds (Mauch Chunk Formation collected near Jim Thorpe, Pa. (40° 52′ N, 75° 43′ W)) since pure discrete components have not yet been found in Bloomsburg material. The magnetization varies little in magnitude or direction up to 650 °C but is lost at 700 °C. The fall in the curve essentially defines the Curie temperature, and it seems clear that the NRM in this case is due predominantly to hematite ($T_c = 675$ °C). However the very small decrease in the range 500 to 600 °C could be due to the presence of a very small amount of magnetite. In *thermally-distributed components* the blocking temperatures range down for hundreds of degrees below $T_c$ so that the demagnetization curve falls regularly. An example is given in Figure 3 (curve 9). The demagnetization curve defines only a lower limit for $T_c$ and therefore provides little information about the chemical nature of the minerals responsible.

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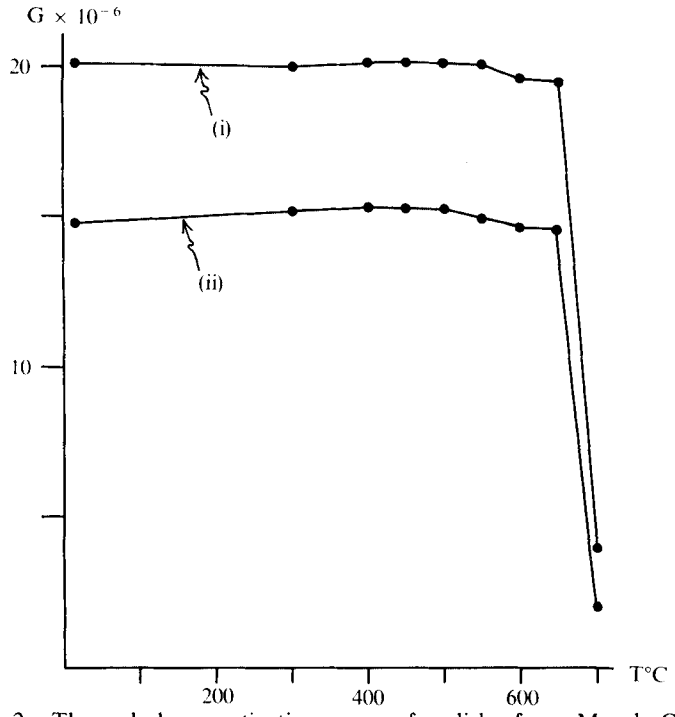



FIG. 2.—Thermal demagnetization curves for disks from Mauch Chunk red beds, showing characteristic thermally-discrete components. Remanent intensity per $cm^3$ is plotted as a function of temperature $T$. The NRM direction for (i) is (292, −38) and after heating to 600 °C this becomes (299, −39) a change of 5°. Values for curve (ii) are respectively (294, −45) and (302, −42) a change of 6°.

## 5. Results of thermal demagnetization

There are three behaviour patterns. *Type A* behaviour is shown by disks whose NRM directions are northerly. The directions show only small changes up to 600 °C (Figure 4). The intensity increases by about 10 per cent in the range 450 to 600 °C and is lost finally between 650 and 700 °C (Figure 3, curve 10). *Type B* behaviour is shown in disks with scattered NRM directions. Upon demagnetization the directions move progressively towards the *Type A* directions reaching there at about 500 °C; there is a large decrease in the divergences between disks from the same and different samples. This remarkable improvement in precision is shown for one sample in Figure 5, and for all disks after heating to 550 °C in Figure 1. The demagnetization curves usually show an increase sometimes by a factor 2 and occasionally by a factor 4. An example for 2 disks from the same sample (directions in Figure 5) is given in Figure 3 (curves 14). *Type C* behaviour is shown by only 1 sample (level 9, 2 disks). The NRM directions are southerly with low inclinations and do not change significantly upon demagnetization. The demagnetization curve falls steadily. There is no northerly-directed high temperature component as occurs in *Type A* and *B* disks but only a southerly directed component of lower thermal stability.

The general increase of intensity after demagnetization at 550 °C is shown in Figure 6. Values, plotted directly, show a skew distribution, which becomes more regular if logarithms are taken. The increases in the means (marked in Figure 6) obtained using the direct and log values are 30 and 36 per cent, respectively.

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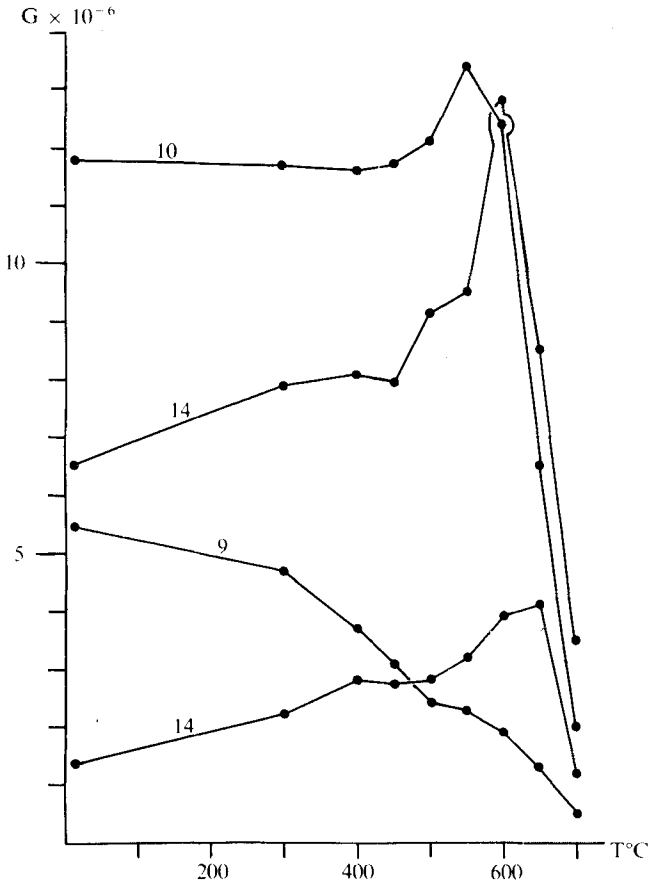



FIG. 3.—Thermal demagnetization curves for Bloomsburg red beds. The remanent intensity per $cm^3$ is plotted as a function of temperature $T$. The curves are labelled with the number of the sampling level (2 from level 14) and the type (see section 5).

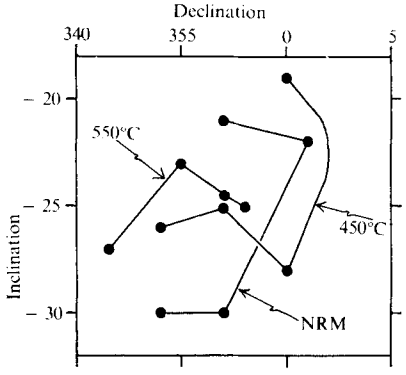



FIG. 4.—Effect of thermal demagnetization on the directions in four disks from the same sample of magnetic *Type A* (curve 10 in Figure 3). Results at three temperatures are given. The directions, which are very well grouped, are plotted on a rectilinear net which occupies only a small part of the sphere of observation (see Figure 5).

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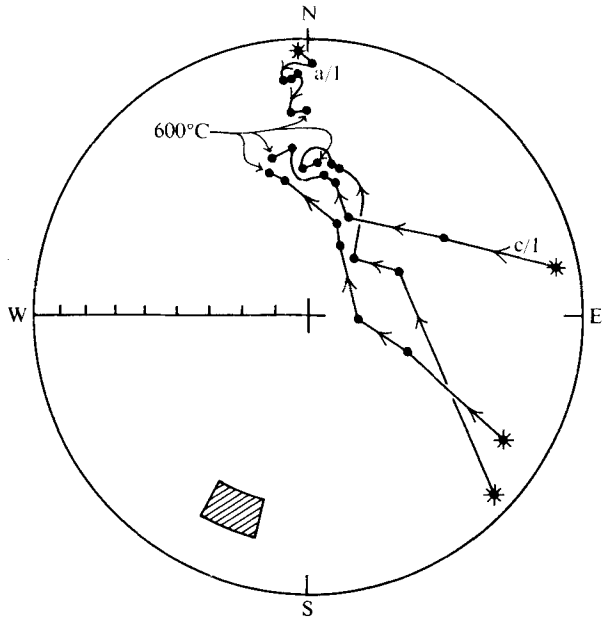



FIG. 5.—Effects of thermal demagnetization on the directions in four disks from the same sample of magnetic *Type B*, level 14. The directions are plotted on the upper hemisphere of an equal-area polar net. Stars indicate the NRM directions. Dots indicate directions after heating to these temperatures—300, 400, 450, 500, 550, and 600 °C the latter being labelled. The links with arrows indicate the trend of the changes. Intensity changes corresponding to these direction changes are plotted for two disks (*a/1* and *c/1*) in Figure 3. The shaded area is the range of Figure 4.

These results are explained by assuming that the NRM is composite being made up of thermally-discrete and thermally-distributed components (Figure 7). The former are directed towards the north with inclination of about −30°, and the latter towards the south with a low negative inclination. The discrete component has a Curie temperature between 650 and 700 °C. The distributed components have blocking temperatures predominantly in the range 300 to 550 °C. In *Type A* disks the discrete components dominate. In *Type B* disks the distributed components are the larger. In *Type C* disks distributed components only occur.

## 6. Thermoremanent magnetization (TRM)

The NRM may contain contributions from all the magnetic phases present or much material may be demagnetized in the natural state. Therefore it is useful to be able to identify the magnetic phases to see how they correspond to those which comprise the NRM. This was done by first giving a TRM to disks in order to induce a magnetization in those phases capable of developing a remanence, and then analysing this remanence thermally. Four disks were heated to 700 °C and cooled in the Earth's field. The ratio NRM/TRM ranges from 0·002 to 0·036. The demagnetization curves fall fairly steadily up to 600 °C to values between 0·5 and 0·05 of the total TRM, the ratio being higher in the *Type A* than in the *Type B* disks (Figure 8). In the range 600 to 650 °C there remains a component of very high stability which is unchanged by repeat heatings. The directions of magnetization are constant within the accuracy of the determinations throughout

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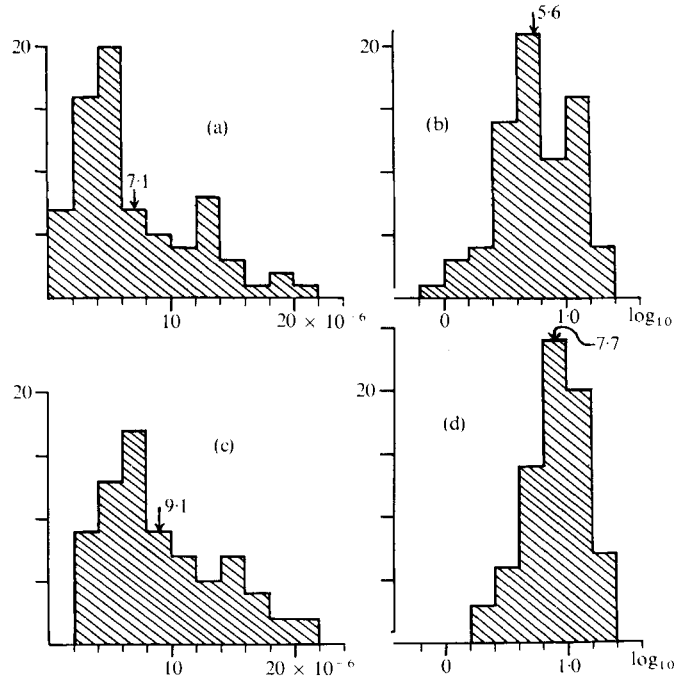



FIG. 6.—Histograms of volume intensities ($G/cm^3 \times 10^{-6}$) of 74 disks, (a) NRM, (b) $\log_{10}$ NRM, (c) after heating to 550 °C, and (d) $\log_{10}$ after heating to 550 °C. The labelled arrows give the means ($G \times 10^{-6}$).

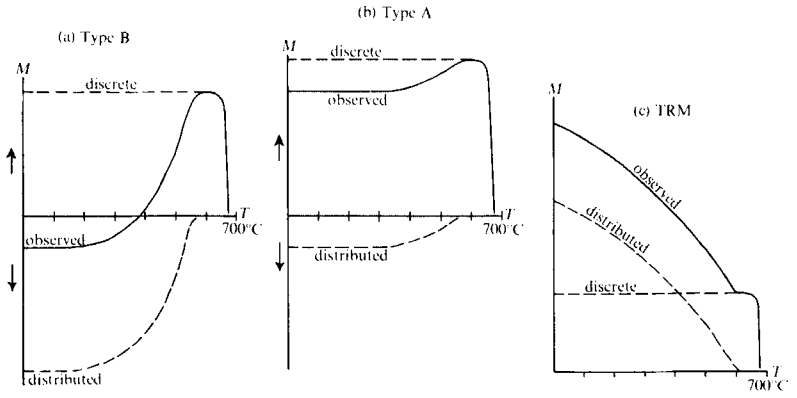



FIG. 7.—Diagram explaining demagnetization results in terms of thermally-discrete and distributed components. Remanence $M$ is plotted as a function of temperature $T$. Broken line represents the hypothetical components (which for the sake of simplicity are assumed antiparallel in (a) and (b)) and full line the smoothed observed remanence curves (cf. Figures 2, 3 and 8).

the range 20 to 650 °C. These properties also may be explained in terms of thermally-discrete and distributed components present in varying proportion (Figure 7). Both components are accurately parallel to one another in contrast to the NRM in which they are roughly antiparallel. It seems therefore that the components behave independently, and are not coupled magnetically, but arise from physically separated phases.

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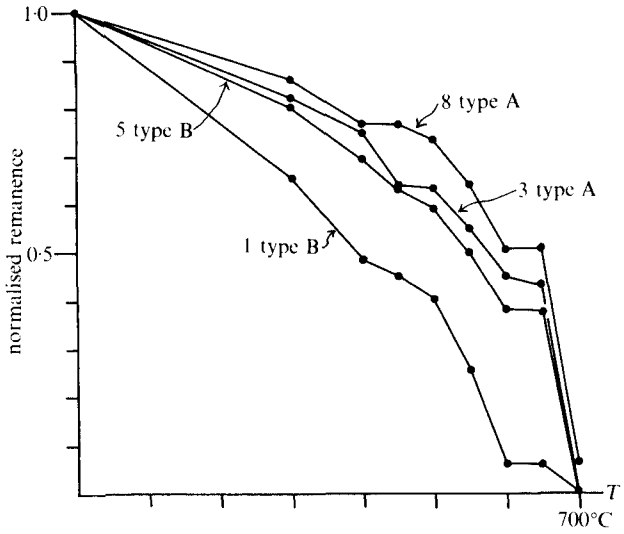



FIG. 8.—Demagnetization curves of TRM. The curves, which are normalized to unit intensity at room temperature, are labelled with the sampling level and type (see Table 1).

**Table 1**

*TRM and NRM in four disks*

The total TRM is that acquired in a field of 0·6 Oe. with an inclination of −65°. $NRM_{550}$ ($TRM_{600}$) is the NRM (TRM) remaining after heating to 550 °C (600) and cooling in zero field. $I$ is the inclination of the remanence, (1) bedding plane vertical, (2) bedding horizontal.

| Level | Type | NRM $G \times 10^6$ | $NRM_{550}$ $G \times 10^6$ | Total TRM $G \times 10^6$ | $I°$ (total TRM) | $\frac{TRM_{600}}{\text{total TRM}}$ | $I°(TRM_{600})$ |
|---|---|---|---|---|---|---|---|
| 3 | A | 4·6 | 6·8 | 588 | −66° (1) | 0·45 | −66° |
| 8 | A | 18·7 | 20·3 | 516 | −65° (1) | 0·51 | −64° |
| 1 | B | 4·8 | 7·2 | 2 147 | −60° (2) | 0·38 | −60° |
| 5 | B | 5·0 | 4·8 | 532 | −61° (2) | 0·06 | −58° |

During acquisition of the total TRM two disks were placed with their bedding planes horizontal, and two with their bedding vertical. In the latter case the inclinations agreed with that of the applied field, but in the former the inclinations were 4 to 5° lower (Table 1). The declinations always agreed with the direction of the horizontal component of the field. The rock, therefore, is slightly anisotropic with respect to TRM, there being an easy direction of magnetization in the bedding plane. The directions were accurately maintained upon demagnetization up to 650 °C so that the anisotropy affects both discrete and distributed components of the TRM; there was no detectable tendency for the more stable components to be less affected by this anisotropy than the less stable components.

## 7. Origin of NRM

Discrete components are characteristic of the TRM or chemical remanent magnetization of crystalline iron oxide material (cf. Roquet 1954, Figure 22) and it is considered reasonable to ascribe them to the crystalline specular ore particles

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which may be detrital in origin and which may have inherited their remanence from an earlier cycle. Thermally-distributed components could arise from an isothermal remanent magnetization in crystalline material (cf. Roquet 1954) but this is very improbable in this case since the TRM also shows this property (Figure 8). Distributed components could also arise from a thermoremanent, viscoremanent, or chemical remanent magnetization of very finely divided material (material which geologically would be classified as "amorphous") whose particle diameter is less than the critical blocking diameter for the time scale and temperatures operative during the magnetization process. The critical diameters for hematite for relevant conditions are less than 1 $\mu$ (Chevallier 1951, Creer 1957, 1961). It is therefore considered reasonable to ascribe the distributed components to the red staining material. This identification of discrete components with the black crystalline particles and the distributed components with the "amorphous" red staining is tentative only.

The difference in the directions of the distributed and discrete components suggests that they were formed at different times. In Figure 1 the direction of the Permian field, as deduced from the Permian data obtained elsewhere in the U.S.A., is plotted; components in this general direction and of variable magnitude could produce the scattered directions of NRM observed. The very close correlation of the presumed Permian field direction and the directions of disks showing *Type C* behaviour (pure distributed components) is notable. The suggestion is therefore that the distributed components are of Permian age.

Two possible origins of the thermally-distributed components are considered qualitatively. Firstly, they may be of viscous origin acquired as a result of moderate heating during burial associated with the Appalachian orogeny. Subsequent uplift and lowering of temperature rendered the magnetization stable. A tentative application of Neel's theory (Neel 1955) to this situation (a rigorous application requires a knowledge of the crystal magnetic properties and is clearly impossible at present) suggests that heating to 100 °C for time of the order of a million years in the Earth's field would be sufficient to induce a distributed component which is removed at 550 °C in the laboratory (time scale of minutes). The second possibility is that the distributed components are of chemical origin (Doell 1956). They could, for instance, have originated as a result of a process of nucleation, perhaps facilitated by moderate heating during burial, so that the red staining particles exceed the critical blocking diameter for those particular conditions and thus acquired a staple remanence in the direction of the prevailing field.

The thermally-discrete components are of exceedingly high magnetic and thermal stability. It probably predates the distributed component, and, if the suggestion as to the age of the latter is correct, it is older than Permian. The uniformity of directions (Figure 1(b)) in rocks showing much penecontemporaneous deformation suggests that the discrete components were acquired after deposition. Such a remanence may have been built up either by bodily rotation, prior to compaction, of the black crystalline grains (Irving 1957) allowing their alignment in the prevailing field direction, or by chemical change of the grains. There is no evidence for such chemical changes, and it is possible, therefore, that the discrete component provides a record of the geomagnetic field direction in Upper Silurian time. The usefulness of the component in this regard may be affected by anisotropy; if the anisotropy was acquired concurrently with the magnetization process then there may be no effect, if it was acquired prior to the acquisition of remanence then a deflection of the discrete remanence may be

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expected. At all events it is most unlikely, because of the geometry of the anisotropy, that the declination will have been affected, and the maximum effect on inclination is not likely to exceed 5°. The answers to many of the problems will only become clear when many studies of this type are carried out on Silurian beds throughout the Appalachians. In the meantime it is not claimed that the directions of the discrete component in Figure 1(b) are necessarily those of the geomagnetic field in this region in the Upper Silurian, but only that the evidence available at present is consistent with that hypothesis. Certainly the physical stability of the discrete component is as high as any yet observed in rocks.

**Table 2**

*Silurian results from the Appalachians*

$D_m$, $I_m$ declination and inclination of the mean direction; $k$ is the estimate of Fisher's $\kappa$ and $a_{95}$ is the circle of confidence ($P = 0{\cdot}05$) giving unit weight to sampling levels (1) and specimens (2). $\phi$ is the angle between $D_m$ and the strike, and $dD_m$ is the error ($P = 0{\cdot}05$) in $D_m$; $dD_m$ value is not given in the last entry since the error $a_{95}$ is probably underestimated.

| Formation and age | $D°_m$, $I°_m$ | $k$ | $a°_{95}$ | strike °E of N | $\phi° \pm dD_m$ | references |
|---|---|---|---|---|---|---|
| Bloomsburg, Cayugan (Su) | 354, $-32\frac{1}{2}$ | $62^{(1)}$ | $5^{(1)}$ | 68 | $74 \pm 6$ | herein |
| Rosehill, Lr. Niagaran (Sm) | $324\frac{1}{2}$, $-39$ | $33^{(1)}$ | $13\frac{1}{2}^{(1)}$ | 28 | $63\frac{1}{2} \pm 17\frac{1}{2}$ | Graham (1949) |
| Clinton iron ore, Lr. Niagaran (Sm) | 143, $+19\frac{1}{2}$ | $107^{(2)}$ | $11\frac{1}{2}^{(2)}$ | 24 | 61 | Howell & others 1958 |

## 8. Statistical analysis

Fisher's method is used (Fisher 1953). Disk directions after thermal cleaning at 550 °C were combined giving each unit weight to obtain the means for each hand sample. Where two samples were obtained from the same level their means were combined giving each unit weight. The mean direction for level 9 (*Type C*) is (187, −8). The means for the levels showing *Type A* and *B* behaviour are plotted in Figure 9; these means were combined giving each unit weight to obtain the overall mean direction ($D_m$, $I_m$) for the high temperature components (Table 2). For 17 levels the resultant is 16·74 and the circular standard deviation 10°. This is a most conservative weighting procedure so that the computed error, if anything, underestimates the accuracy with which the mean is defined.

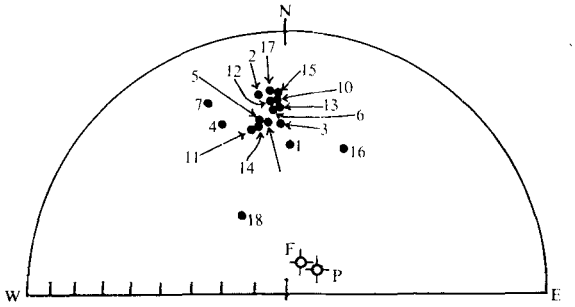



FIG. 9.—Mean directions at 17 sampling levels after partial thermal demagnetization at 550 °C. (Legend as for Figure 1.)

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In Table 2, this result is compared against previous data from the Valley and Ridge Province. The first of these (Graham 1949) is based on 35 samples collected from five sites in the Rosehill Formation near Cumberland (Md) and Berkeley Springs (Va) (Figure 10). In the former locality a fold test demonstrates the stability of the magnetization. The Rosehill data were presented graphically by Graham (it was obtained prior to the development of satisfactory statistical techniques in palaeomagnetism) and in Table 2 a numerical analysis of his plots is given, according unit weight to each site ($N = 5$, $R = 4{\cdot}85$). The inclination does not differ significantly from that of the Bloomsburg, but the declinations differ by $29\frac{1}{2}°$ which exceeds the combined declination errors ($dD_m = a_{95} \cdot \sec I_m$). A test (Watson 1956) of the hypothesis that the mean directions of the Rosehill and Bloomsburg are identical may be made using the statistic

$$\frac{2(N-p)}{2(p-1)} \frac{\sum R - R_T}{N - \sum R}$$

where $p$ is the number of formations (in this case 2), $N$ is the total number of sites (22), $R$ are the resultants of the site means for each rock formation, and $R_T$ is the resultant of all sites lumped together (21·62). The statistic has the value 9·4 which when referred to the $F$-ratio tables with $2(p-1)$ and $2(N-p)$ degrees of freedom is significant at $P = 0{\cdot}5$. It is clear therefore that there is a real difference between the means of the Bloomsburg and Rosehill beds and that this difference is predominantly in declination. However the angle ($\phi$) which the horizontal components make with the strike in each case is constant at the present statistical levels.

A third result is available from a single site in the Clinton iron ore of Alabama (Howell, Martinez & Statham 1958). It is based on only 7 samples (16 specimens) and no stability evidence other than the departure from the present field. The directions are approximately reversed from those observed farther north. The horizontal component is parallel (within the errors) to those in the Rosehill Formation (Figure 11) and bears the same relation to the strike, but little reliance can be placed on this data without further stability studies and the result is not considered further.

## 9. Discussion

The declination discrepancy between the Rosehill and Bloomsburg Formations is discussed under three headings.

### 9.1. *Time variations in the field*

The Rosehill and Clinton beds are Middle Silurian whereas the Bloomsburg beds are Upper Silurian. Therefore it is possible that the discrepancy arose from changes in the field direction between Middle and Upper Silurian.

### 9.2. *Stress effects*

It is conceivable that the remanence is stress controlled, but this is considered unlikely for two reasons. Firstly, such effects, if they occurred, are hardly likely to have confined themselves to one direction component only. Secondly, the beds have undergone folding about horizontal axes and correction for this in the Rosehill Formation brings the directions into good agreement; this result virtually eliminates any possibility of tectonic stress effects in this case, but clearly more studies of beds tilted in different directions are needed.

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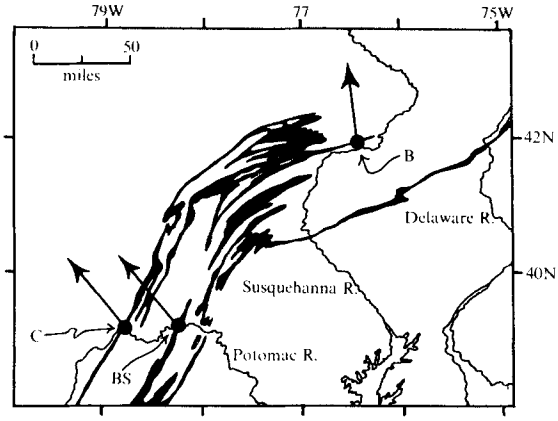



FIG. 10.—Sketch map of the central Appalachians. The outcrop of Silurian beds is coloured black. B is Bloomsburg, C Cumberland, and BS Berkeley Springs. The arrows show the directions of the horizontal component of magnetization.

9.3. *Bending in plan*

The directions in Figures 10 and 11 bring to mind Carey's orocline hypothesis. The palaeomagnetic directions are consistent with the view that in Silurian time the central Appalachian geosyncline was straight and that it has since undergone bending by 40° about a hinge at approximately 41 °N, 77·5 °W. It seems reasonable to follow Carey's practice of naming oroclines, and the name *Susquehanna orocline* is proposed. It is stressed that all that is said here is that the palaeomagnetic observations available at present are consistent with this change in trend being an orocline, but they do not of course prove that such is the case. For example, it is possible that the divergence observed is of very local significance due, say, to tectonic rotations in the immediate vicinity of the sampling localities, and that the results have no bearing on the origin of the change in trend of the Valley and Ridge Province as a whole at this point. However, it is worthwhile formulating a specific hypothesis because it can be tested rigorously by further palaeomagnetic

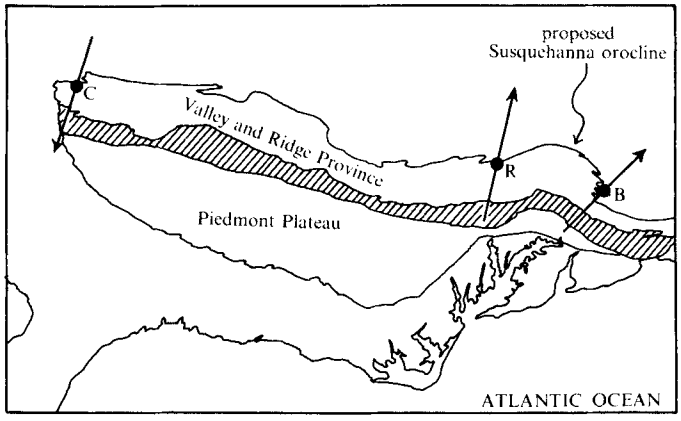



FIG. 11.—Sketch of central and southern Appalachians showing tectonic provinces. The shaded strip is the Blue Ridge province with Triassic Basins in the north. The arrows show the directions of the horizontal component of magnetization; B Bloomsburg, R Rosehill, and C Clinton iron ore.

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work. If future observations show that the directions in rocks of the same age are the same along the whole length of the Appalachian chain, and this is demonstrated in beds of all the ages present, then clearly the application of the orocline hypothesis to this mountain system can be discounted. On the other hand if a constant relation between direction and tectonic trend is found, then the case for the hypothesis would be substantial.

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